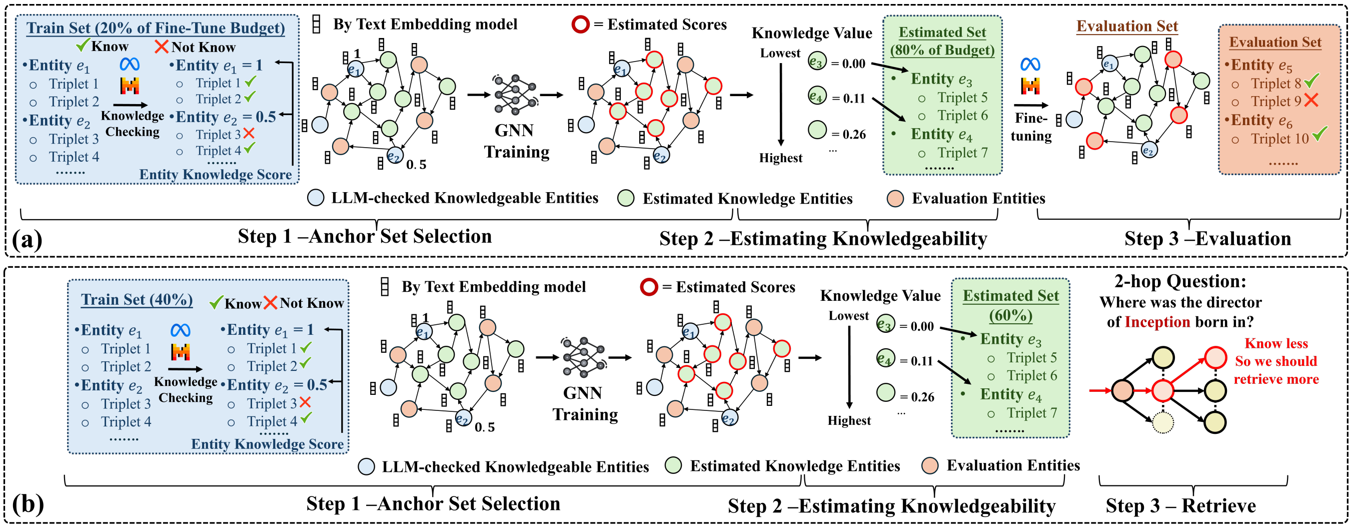

Figure 4: Homophily-guided Knowledge Injection and Retrieval: The process begins by training a GNN on a subset of entities with ground-truth knowledgeability scores (Blue Nodes) obtained by querying the base LLM. The trained GNN then infers the knowledgeability scores for all remaining entities (Green Nodes). Based on these predictions, triplets associated with entities estimated to have the lowest knowledge values are selected until the budget is met. Finally, the base LLM is fine-tuned on these less-known triplets to efficiently inject new knowledge, and its improved performance is measured on a held-out test set (Orange Nodes) in Figure 4(a). The estimated knowledgeability scores also guide retrieval, as illustrated in Figure 4(b).

A Appendix

A.1 Experimental Setting Visualization

A.1.1 Homophily-guided Knowledge Injection. Figure 4 (a) illustrates the pipeline of homophily-guided knowledge injection. In general, we leverage knowledge homophily to train a GNN estimator that predicts entity knowledgeability, and then use these estimated knowledgeability scores to identify less-known triplets for fine-tuning LLMs. The global procedure is as follows:

- **Step 1 - Fine-tuning Stage - Anchor Set Selection**: The process begins with a predefined triplet budget for fine-tuning LLMs. From this budget, 20% is allocated to anchor triplets, while the remaining 80% is reserved for knowledgeability estimation. Anchor triplets are constructed using an entity-centric sampling strategy: entities are randomly selected one by one, and all their associated triplets are added until the 20% quota is met. These anchor entities are used for training the GNN knowledge estimator and are shown as the blue "Known Knowledge Scores" nodes in Figure 4(a). Their ground-truth knowledgeability is obtained by querying the base LLM on their anchor triplets and aggregating the outcomes (see Section 3.1). With knowledge homophily, the GNN is then trained on these anchor entities to learn the relation between graph topology and knowledgeability.
- **Step 2 - Fine-tuning Stage - Estimating Knowledgeability of Remaining Set**: After training, the GNN estimator infers scores for all unlabeled entities (i.e., those with unknown knowledgeability). Entities are then ranked by their predicted knowledge value, $\mathcal{K}(v)$, where lower scores indicate a higher unknown level to the LLM. Finally, triplets linked to the least knowledgeable entities are selected as the remaining 80% of fine-tuning budget.
- **Step 3 - Evaluation Stage**: The selected triplets are combined with the initial anchor set to form the fine-tuning dataset. This dataset, enriched with facts less known to the LLM, is used for fine-tuning. We evaluate the procedure in two ways. First, we measure selection quality, verifying whether the selected triplets are indeed unknown to the LLM. Second, we sample 2% of triplets (orange nodes) that are neither used in fine-tuning nor involve entities overlapped with fine-tuned triplets, to assess the generalization gain. As shown in Table 1, the fine-tuned LLM exhibits clear improvements in both selection quality and generalization performance. These results demonstrate the effectiveness of our homophily-aware knowledge injection in identifying knowledge deficiencies of LLMs and maximizing fine-tuning gains.

A.1.2 Homophily-guided Knowledge Retrieval. Figure 4(b) details the operational pipeline of our homophily-guided knowledge retrieval method in Section 4, designed to enhance the quality of the retrieved context to further improve multi-hop question answering.

The process begins with the creation of a multi-hop question set from 2-hop and 3-hop triplet paths. To ensure a fair evaluation, the entities that constitute these question paths are explicitly excluded from the GNN training data to prevent data leakage. From the remaining pool of entities, 40% are sampled to train the GNN regressor. This trained model infers the knowledge scores, $\mathcal{K}(v)$, for other entities, quantifying the awareness of LLMs of these other entities. For a given multi-hop question, the retrieval process commences with entity linking to anchor the query to a starting entity in the graph. From this point, a GNN-based Beam Search (G-BS) is employed to explore potential reasoning paths. The core of this method is its unique scoring function, $\mathcal{S} \times (1 - \alpha \cdot \mathcal{K}(u))$, which balances semantic relevance with a penalty based on the knowledgeability of the next entity ($\mathcal{K}(u)$). By penalizing the expansions toward well-known entities, the search prioritizes retrieving facts

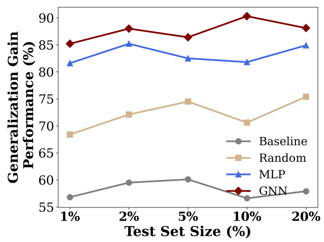

Figure 5: The knowledge injection performance of the fine-tuned Mistral models on the CoDEx-S dataset across varying test set sizes. The GNN-guided approach maintains a significant performance advantage over other methods.

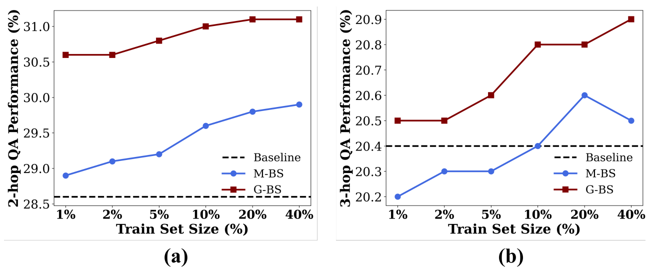

Figure 6: The knowledge-aware retrieval performance across the varying training budgets of the underlying knowledgeability estimator. For both 2-hop (left) and 3-hop (right) QA on the CoDEx-S dataset, the GNN-based search (G-BS) consistently outperforms the homophily-agnostic MLP-based search (M-BS) and the semantic baseline.

Table 3: Correlation results between entity knowledgeability scores under Full and Sparse settings across datasets.

Dataset	Full vs Sparse75%		Full vs Sparse50%	
	Pearson	Spearman	Pearson	Spearman
CoDEx-S	0.9577	0.9484	0.8490	0.8398
MVPKG	0.9376	0.9357	0.8839	0.8775
PharmKG	0.9444	0.9408	0.8773	0.8650
T-Rex	0.9177	0.9156	0.7957	0.7834
WD50K	0.9370	0.9278	0.8275	0.8080

with higher information gain, thereby providing the LLM with specific context to answer the query.

A.2 Additional Results

A.2.1 Knowledgeability Score Robustness. To validate that the calculated entity knowledgeability score ($\mathcal{K}(v)$) is robust to our triplet sampling algorithm, we conducted an experiment to assess its stability under data sparsity. For each knowledge graph, we created two sparsified versions: one retaining 75% of the original triplets (Sparse75%) and another retaining 50% (Sparse50%). We then recalculated the knowledgeability scores for all entities on these sparse graphs and computed the Pearson/Spearman correlations against the scores derived from the full, original graph. The results, presented in Table 3, demonstrate the stability of our metric.

For the Sparse75% condition, the Pearson correlation consistently exceeded 0.91 across all datasets, indicating a very strong linear relationship. Even with half of the relational data removed in the Sparse50% condition, the scores maintained a strong correlation with the originals, with Pearson values generally above 0.80. The consistently high values for both Pearson and Spearman coefficients confirm that the entity knowledgeability score is not simply a byproduct of local graph density. These findings provide strong empirical evidence that our metric captures a stable property of the LLM's knowledge, which can be reliably estimated even from an incomplete set of relational facts.

A.2.2 Sensitivity Analysis: Knowledge Injection. To assess the robustness of our findings and ensure our conclusions are not dependent upon a specific test set size in the knowledge injection experiment, we conduct a sensitivity analysis. This analysis evaluates the performance of our fine-tuned models on the CoDEx-S dataset using the Mistral-7B model, mirroring the primary experiment in Section 4.2. We constructed five randomly sampled test sets, each representing a different proportion of the total dataset: 1%, 2%, 5%, 10%, and 20%. It was ensured that none of the entities used to train the GNN/MLP knowledgeability estimator appeared in any of the test sets. The results of this analysis are presented in the Figure 5. Our key findings of this analysis are:

- **Consistent Superiority:** The GNN-guided knowledge injection method consistently outperforms the MLP, Random, and Baseline approaches across all evaluation set sizes. This confirms the significant advantage of leveraging knowledge homophily.
- **Performance Stability:** Although minor fluctuations were observed, attributable to statistical variance in sampling, the performance of all methods remained relatively stable across evaluation set sizes. This finding suggests that the measured performance of the models is not merely a statistical artifact of a particular test set size.

A.2.3 Sensitivity Analysis: Knowledge Retrieval. To evaluate the robustness of our knowledge-aware beam search method (G-BS and M-BS), we conducted a sensitivity analysis on the amount of training data used for the knowledgeability estimators. The primary experiment in our paper utilizes GNN and MLP models trained on 40% of the available entities. This analysis investigates how performance on the multi-hop question-answering task varies when this training budget is reduced. For this sensitivity analysis, we select the CoDEx-S dataset. We trained a series of GNN and MLP knowledgeability estimators on progressively larger subsets of entity data: 1%, 2%, 5%, 10%, 20%, and 40%. Each resulting estimator was then integrated into the G-BS and M-BS retrieval methods, respectively, and evaluated on the fixed set of 1,000 2-hop and 3-hop questions. The performance of the Semantic Beam Search baseline is independent of this training budget and remains constant. The results are presented in the Figure 6 for 2-hop (left) and 3-hop (right) QA performance. Our key findings of this analysis are:

- **Consistent Superiority of G-BS:** The GNN-based approach (G-BS) consistently outperforms both the homophily-agnostic MLP-based method (M-BS) and the baseline across all training set sizes and for both 2-hop and 3-hop questions. This confirms that the advantage of leveraging knowledge homophily is robust, even under data-scarce conditions.

Table 4: Statistics of the original knowledge graph and the sampled largest connected component.

Dataset	# Nodes		# Triplets		# Avg. Deg		# Avg. CC	
	Original	Sampled	Original	Sampled	Original	Sampled	Original	Sampled
T-Rex	3153568	46891	6566790	193781	4.16	8.26	0.1473	0.5170
WD50K	41334	5140	233838	34208	11.31	13.31	0.0996	0.1332
PharmKG8K	7262	6877	479902	98537	132.16	28.65	0.2512	0.0824
MVPKG	137117	9055	1857410	255697	12.46	28.24	0.0013	0.0140
MVPKG w/o t	137117	9055	1857410	116127	12.46	12.82	0.0013	0.0140
CoDEx-S	2034		36543		35.93		0.0952	

- **Performance Scaling with Data:** The performance of both G-BS and M-BS improves as the training budget for the knowledgeability estimator increases. This suggests that more accurately estimated knowledge scores lead to better path retrieval for QA.

A.2.4 Role of Knowledge Graph Probing. Our goal in this work is not to propose a new knowledge-graph (KG) probing method, but to use KG structure as a lens to study how an LLM's factual knowledge is organized topologically, specifically, whether topologically proximate entities exhibit similar knowledgeability ("knowledge homophily"). Because the main scientific claim concerns the presence and utility of such a homophily pattern, the probing procedure is used solely as a measurement instrument to obtain a consistent entity-level knowledgeability signal that can be placed on the KG for analysis. As a result, the specific choice of probing method is not the focus of this work, and we do not claim superiority or equivalence across probing techniques. Our conclusions are based on the observed topological patterns induced by the chosen signal, rather than on properties of the probing method itself.

A.2.5 Scalability and Computational Considerations. While our experiments focus on small to mid-scale knowledge graphs, the computational implications for larger graphs warrant clarification. As we exploit the property of knowledge homophily: topologically close entities tend to exhibit similar knowledgeability, our approach does not require probing all entities or triplets in the graph, unlike prior methods that rely on exhaustive or iterative probing to identify knowledge deficiencies in LLMs [30]. Instead, we probe only a subset of entities, using their observed knowledgeability signals to infer the remaining ones via graph structure. This substantially reduces computational cost, as well as LLM-specific overhead such as inference time and token usage, since fewer prompts are required.

We acknowledge that our method introduces additional components beyond probing, namely the use of a GNN for knowledgeability inference over the graph. However, GNN inference is known to scale linearly with respect to the number of nodes and edges in the graph for a fixed number of layers, which is well established in the literature [12, 34]. Moreover, existing large-scale benchmarks have demonstrated that GNNs can be trained and evaluated on graphs with millions of entities, such as those in the Open Graph Benchmark [9]. In comparison, the knowledge graphs used in our work are considerably smaller and therefore well within the regime where GNN-based inference is computationally feasible.

A.2.6 Datasets. Our experiments are designed to evaluate and compare the knowledgeability of the LLM across multiple datasets. We illustrate our process on five datasets: **MVPKG** (covering U.S. legislative, election, diplomatic data, etc.), **T-Rex** (containing large-scale high-quality alignments between DBpedia abstracts and Wikidata triples), **PharmKG8K** (biomedical knowledge graph), **WD50K** (derived from Wikidata statements), and **CoDEx-S** (extracted from Wikidata and Wikipedia). Table 4 provides the dataset statistics.

- **MVPKG** [21]: The MVPKG dataset encompasses U.S. legislative, election, and diplomatic data as well as conceptual knowledge from Wikidata. It originally contains 1,857,410 triplets, 137,117 entities, and 602 relations. Due to scale considerations, we extract the largest strongly connected component, which comprises 255,697 triplets, 9,055 entities, and 602 relations. The MVPKG dataset had a temporal attribute and was evaluated with the temporal component included and excluded. For each triplet, two prompts are generated (with time and without time). Consequently, each entity in MVPKG is assigned two knowledgeability scores corresponding to the two prompt variants for further analysis of the effect of inclusion of temporal information. All other datasets have only one knowledgeability score due to lack of temporal attributes.
- **T-Rex** [4]: The T-Rex dataset is constructed from Wikipedia abstracts aligned with Wikidata entities in English. It contains 6,566,790 unique triplets; the largest connected component comprises 193,781 triplets, 46,891 entities, and 423 relations.
- **PharmKG8K** [43]: The PharmKG8K multi-relational, attributed biomedical KG, composed of around 500,000 individual interconnections between genes, drugs, and diseases, with 29 relation types over a vocabulary of around 8000 disambiguated entities. Given the scope of the dataset, we used a strongly connected component of 98,537 edges, 6,877 entities, and 29 relations.
- **WD50k** [5]: The WD50K dataset was created using the Wikidata RDF dump of August 2019. It has 233,838 edges and 41,334 entities. Since being extracted from Wikidata, there were 14,858 triplets common between the WD50K dataset and the T-Rex largest connected component selected. These were removed to make sure that common triplets were not overshadowing the result comparison between these datasets. Following that, the largest strongly connected component was selected for experimental purposes. This LCC had 34,208 edges, 5,140 entities, and 193 relations.
- **CoDEx-S** [27]: CoDEx is a collection of knowledge graph completion datasets extracted from Wikidata/Wikipedia, comprising three subsets of varying sizes. We select CoDEx-S due to its high proportion of triplets involving the "occupation" relation, which poses greater challenges for LLMs, since individuals may hold multiple occupations. CoDEx-S contains 36,543 triplets, 2,034 entities, and 42 relations.